

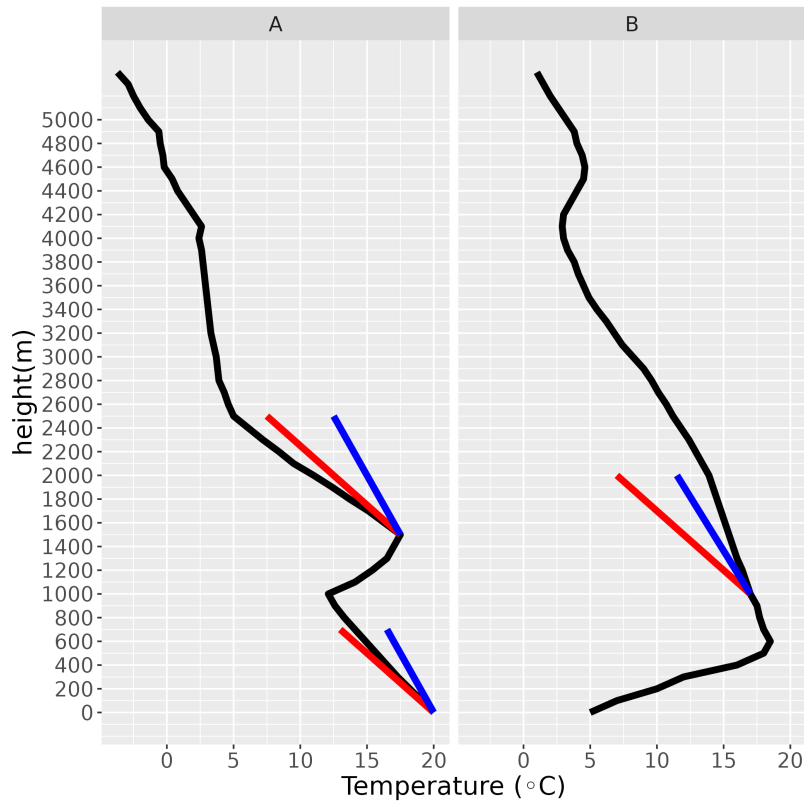


Figure 1: Illustration for Question 5. Each panel is an environmental temperature profile taken above a station in the Northern Cape during winter. Black lines indicate the environmental temperature profile, red lines the dry adiabatic and blue lines the saturated adiabatic lapse rate in the indicated atmospheric layers. In panel A, the adiabatic lapse rates are shown for the 0–700m and the 1500–2500m height layers, for panel B for the 1000–2000m .

For all questions in this section, unless indicated otherwise, assume the dry adiabatic lapse rate is $\Gamma_d = 0.01\text{K} \cdot \text{m}^{-1} = 10^\circ\text{C} \cdot \text{km}^{-1}$ and the saturated adiabatic lapse rate $\Gamma_s = 0.006\text{K} \cdot \text{m}^{-1} = 6^\circ\text{C} \cdot \text{km}^{-1}$. The dew point temperature lapse rate in unsaturated air can be assumed equal to $0.002\text{K} \cdot \text{m}^{-1} = 2^\circ\text{C} \cdot \text{km}^{-1}$.

1. Regarding the profiles illustrated in Figure 1, [total = 8 marks]
 - 1.1 One of the panels represents a night-time observation, the other one a mid-afternoon observation. Which is the night-time one? Explain briefly. (3 marks)
 - 1.2 For the day-time profile, indicate the height at the lower and upper bounds of the inversion marking the entrainment zone above the boundary layer (2 marks).
 - 1.3 In the following three cases, indicate whether the layer indicated is stable, conditionally unstable, neutrally stable with respect to a saturated air parcel, neutrally stable with respect to a dry air parcel or absolutely unstable: (3 marks)
 - (a) The profile A near-surface mixed-layer
 - (b) The 1500–2500m layer in profile A
 - (c) The 2500–3500m layer in profile B

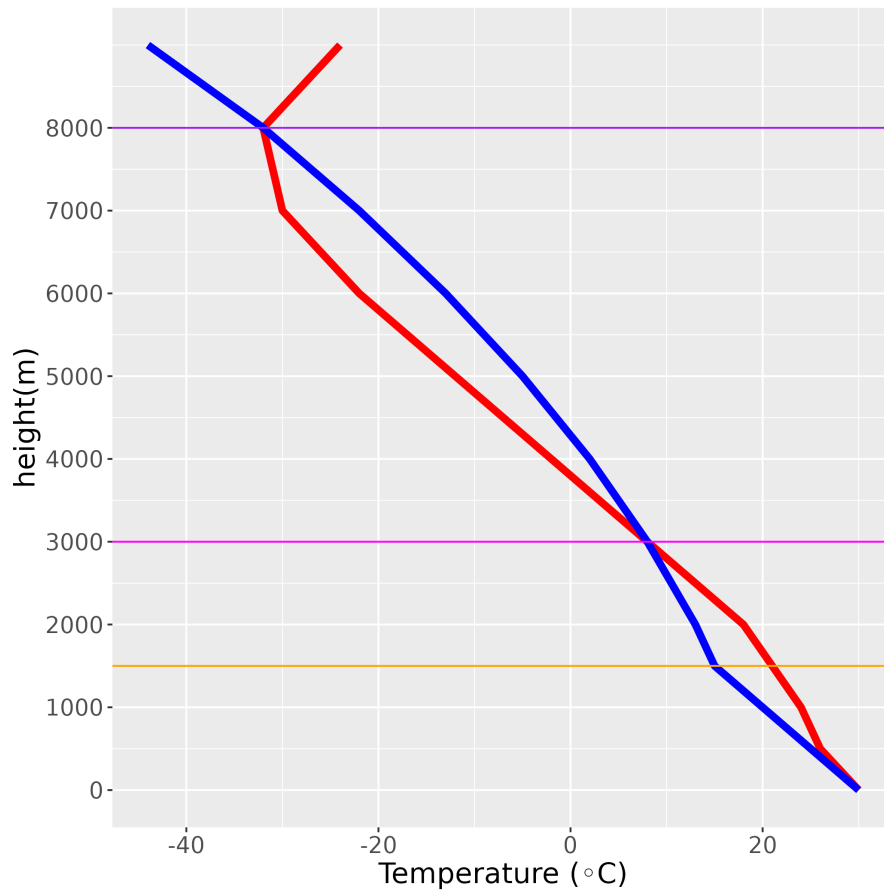


Figure 2: Illustration for Question 4. The blue line is the parcel temperature profile for an ideal adiabatically lifted parcel, with LCL at 1000m and the red line is the environmental lapse rate. Heights of intersection between the two curves are indicated by orange and purple horizontal lines.

2. Regarding the scenario illustrated in Figure 2 [total = 7 marks]

- 2.1 If air were lifted adiabatically from the ground level, at approximately what height would the base of cloud form? (1 mark)
- 2.2 To what height would surface air need to be lifted before it would continue to rise by itself if pushed higher? (1 mark)
- 2.3 If air were lifted from the ground level to just above the level of free convection and then released, at what height would the cloud top be found? (1 mark)
- 2.4 Estimate the dew point temperature at the surface to the nearest °C, showing all working (2 marks).
- 2.5 To calculate convective inhibition (CIN) for this profile, what would the upper bound for the layer over which the calculation is performed be? (1 mark)
- 2.6 To calculate convective available potential energy (CAPE) for this profile, what would the upper bound for the layer over which the calculation is performed be? (1 mark)

3. Moist Thermodynamics [15 marks] :

3.1 In each of the scenarios below, indicate whether the described atmospheric layer is stable, neutrally stable with respect to a dry air parcel, neutrally stable with respect to a saturated air parcel, conditionally unstable or absolutely unstable. Assume fixed values for the dry adiabatic lapse rate, $\Gamma_d = 0.01\text{K} \cdot \text{m}^{-1} = 10^\circ\text{C} \cdot \text{km}^{-1}$ and the saturated adiabatic lapse rate $\Gamma_s = 0.006\text{K} \cdot \text{m}^{-1} = 6^\circ\text{C} \cdot \text{km}^{-1}$. [total = 8 marks]:

- (a) A layer where $\Gamma_e = 0.005\text{K} \cdot \text{m}^{-1}$ (1 mark)
- (b) A layer where the environmental temperature decreases linearly from $T_e = 26^\circ\text{C}$ at the surface to $T_e = 14^\circ\text{C}$ at $z = 1000\text{m}$ (1 mark)
- (c) A 1km deep inversion layer (1 mark)
- (d) A shallow, foggy inversion layer (1 mark)
- (e) A thoroughly mixed, unsaturated lower atmospheric boundary layer above a dark surface experiencing intense solar heating (1 mark)
- (f) A layer in which equivalent potential temperature (Θ_{eq}) is constant with 12% relative humidity throughout (1 mark)
- (g) A layer in which equivalent potential temperature (Θ_{eq}) is constant with height with 100% relative humidity throughout (1 mark)
- (h) A layer in which the environmental lapse rate Γ_e varies between $0.008\text{K} \cdot \text{m}^{-1}$ and $0.009\text{K} \cdot \text{m}^{-1}$. (1 mark).

3.2 Consider the following five atmospheric profiles A, B, C, D and E, all starting from the surface at sea level with surface pressure $P = 1000\text{hPa}$, characterised as follows (where CAPE is convective available potential energy; CIN is convective inhibition; LCL is the lifting condensation level; and EL is the equilibrium level). Provide only the letter corresponding to the correct profile [7 marks]:

A CAPE = $2000\text{J} \cdot \text{kg}^{-1}$; CIN = $1500\text{J} \cdot \text{kg}^{-1}$; LCL = 830hPa; EL = 130hPa

B CAPE = $1000\text{J} \cdot \text{kg}^{-1}$; CIN = $400\text{J} \cdot \text{kg}^{-1}$; LCL = 920hPa; EL = 440hPa

C CAPE = $100\text{J} \cdot \text{kg}^{-1}$; CIN = $700\text{J} \cdot \text{kg}^{-1}$; LCL = 970hPa; EL = 680hPa

D CAPE = $250\text{J} \cdot \text{kg}^{-1}$; CIN = $2400\text{J} \cdot \text{kg}^{-1}$; LCL = 870hPa; EL = 300hPa

E CAPE = $1500\text{J} \cdot \text{kg}^{-1}$; CIN = $750\text{J} \cdot \text{kg}^{-1}$; LCL = 885hPa; EL = 200hPa

- (a) In which of the cases is a thunderstorm most likely to develop? (1 mark)
- (b) In which of the cases is a thunderstorm least likely to develop? (1 mark)
- (c) In which of the cases would you expect to see the fastest updrafts if thunderstorms were to develop in all cases? (1 mark)
- (d) In which of the cases would you expect the cloud base to be closest to the surface, if clouds were to form in all cases? (1 mark)
- (e) In which of the cases would you expect the coldest cloud top temperatures, if convective clouds were to form in all cases? (1 mark)
- (f) In which of the cases would you expect the lowest cloud top height, if convective clouds were to form in all cases? (1 mark)
- (g) In which of the cases is hail most likely to form, if thunderstorms develop in all cases? (1 mark)